

# Delayed Verification in Multi-Agent Factuality Dynamics: Stability and Optimal Corrector Placement

(formal skeleton — v0)

2026-06-21

**One-line thesis.** In a network of LLM agents that exchange beliefs and are re-grounded by *corrector (rectifier) agents* subject to a verification delay  $\delta$ , the closed-loop factuality dynamics reduce, via the *grounded Laplacian*, to a family of scalar delay recurrences. This yields (i) a closed-form stability criterion exposing a *pinning-delay tradeoff* (“verification has an optimal dose”), and (ii) a clean placement problem decoupled into a stability lever and a steady-state-accuracy lever.

## 1 Model

Let  $G = (V, E)$  be a graph on  $V = \{1, \dots, n\}$  with symmetric Laplacian  $L = D - A \succeq 0$ . Each agent  $i$  holds a scalar belief  $b_{i,t} \in \mathbb{R}$  about a claim whose ground truth is  $b^*$ ; write the error  $e_{i,t} = b_{i,t} - b^*$ .

**Three roles.**

- **Corrector / rectifier set**  $R \subseteq V$ ,  $|R| = m$ : nodes with access to verified external evidence, hard-grounded to truth,  $e_{i,t} = 0$  for  $i \in R$ .
- **Free set**  $V_f = V \setminus R$ ,  $n_f = n - m$ , with error vector  $e_t \in \mathbb{R}^{n_f}$ .
- **Faulty set**  $F \subseteq V$ : agents injecting a constant bias toward a false attractor; their net effect on the free nodes is a constant forcing  $g \in \mathbb{R}^{n_f}$ .

**Dynamics.** Free agents run consensus (step size  $\eta > 0$ ) plus a uniform *delayed self-verification* of gain  $\kappa > 0$  and integer latency  $\delta \geq 1$  (the verification protocol re-checks a  $\delta$ -old state against evidence):

$$e_{t+1} = (I - \eta L_g) e_t - \eta \kappa e_{t-\delta} + \eta g, \quad e_t \in \mathbb{R}^{n_f}, \quad (1)$$

where  $L_g := L[V_f, V_f]$  is the principal submatrix of  $L$  on the free nodes — the *grounded Laplacian*.

**Assumption 1** (grounding reachability). Every connected component of the subgraph induced by  $V_f$  contains a node adjacent to  $R$ . Then  $L_g \succ 0$  (standard;  $L_g$  is a symmetric, diagonally dominant  $M$ -matrix with at least one strictly dominant row per component).

**Remark 1** (modeling choices, deliberately minimal). Soft/Byzantine faults, edge-level (message-filtering) correctors, nonlinear belief updates, and a separate gossip delay  $d$  are all deferred. The present model keeps the single spine: *linear consensus + grounded correctors + one verification delay*.

## 2 Reduction to scalar delay recurrences

**Lemma 1** (grounded-Laplacian decoupling). *Let  $L_g = Q \operatorname{diag}(\mu_1, \dots, \mu_{n_f}) Q^\top$  be the spectral decomposition ( $Q$  orthogonal,  $0 < \mu_1 \leq \dots \leq \mu_{n_f}$  by Assumption 1). Put  $x_t = Q^\top e_t$  and  $\hat{g} = Q^\top g$ . Because  $\kappa I$  is scalar (hence commutes with  $L_g$ ), (1) decouples into  $n_f$  independent scalar recurrences*

$$x_{i,t+1} = a_i x_{i,t} - \eta \kappa x_{i,t-\delta} + \eta \hat{g}_i, \quad a_i := 1 - \eta \mu_i. \quad (2)$$

*Proof sketch.* Multiply (1) by  $Q^\top$ ;  $Q^\top(I - \eta L_g)Q = \operatorname{diag}(a_i)$  and  $Q^\top(\kappa I)Q = \kappa I$ .  $\square$

**Scalar stability region.** Mode  $i$  is governed by  $x_{t+1} - a_i x_t + \beta x_{t-\delta} = 0$  with  $\beta = \eta \kappa$ . Let  $\mathcal{D}_\delta \subset \mathbb{R}^2$  be the set of  $(a, \beta)$  for which every root of  $z^{\delta+1} - a z^\delta + \beta = 0$  lies in the open unit disk.  $\mathcal{D}_\delta$  is given in closed form by Kuruklis (1994). The case  $\delta = 1$  is the Jury triangle:

**Proposition 1** (explicit  $\delta = 1$  region). *For  $\delta = 1$ , mode (2) is asymptotically stable iff*

$$\boxed{\eta \kappa < 1 \quad \text{and} \quad \mu_i < \frac{2}{\eta} + \kappa.} \quad (3)$$

*Proof sketch.* Characteristic polynomial  $z^2 - a_i z + \eta \kappa$  with  $a_i = 1 - \eta \mu_i$ . Jury conditions for  $z^2 + c_1 z + c_0$  (roots in unit disk):  $1 + c_1 + c_0 > 0$ ,  $1 - c_1 + c_0 > 0$ ,  $|c_0| < 1$ , here  $c_1 = -a_i$ ,  $c_0 = \eta \kappa$ . The first is  $\eta(\mu_i + \kappa) > 0$  (automatic), the second is  $\mu_i < 2/\eta + \kappa$ , the third is  $\eta \kappa < 1$ .  $\square$

**Remark 2** (delay shrinks the dose ceiling; binding mode is the slowest). For general  $\delta$  the admissible band of  $\beta = \eta \kappa$  in  $\mathcal{D}_\delta$  depends on  $a$  and shrinks as  $\delta$  grows (Kuruklis). The per-mode dose limit  $\kappa_{\max}(a, \delta)$  satisfies  $\kappa_{\max}(a, 1) = 1/\eta$  and is *strictly decreasing* in  $\delta$  for every  $a \neq 0$ , while the degenerate mode  $a = 0$  ( $\mu = 1/\eta$ ) has  $\kappa_{\max}(0, \delta) = 1/\eta$  for all  $\delta$  (there  $z^{\delta+1} = -\eta \kappa$ , so  $|z| = (\eta \kappa)^{1/(\delta+1)}$ ). Numerically (e.g.  $\eta = 0.1$ ):  $a = 0.9$  gives  $\kappa_{\max} = 10, 6.47, 4.86, 3.94, 3.36, 2.96$  for  $\delta = 1, \dots, 6$ ;  $a = 0.5$  gives  $10, 7.81, 6.82, \dots$ ;  $a = 0$  stays at 10. Hence the *binding* dose constraint for the network is set by the *slowest grounded mode* ( $a = 1 - \eta \mu_{\min}$ , closest to 1):  $\kappa < \kappa_{\max}(1 - \eta \mu_{\min}(L_g), \delta)$ . Since grounding more nodes raises  $\mu_{\min}(L_g)$  (interlacing), placement *relaxes* the delay-induced dose limit — coupling Part-I stability to Part-II placement. All four claims (decoupling, the  $\delta=1$  boundary (4), this shrinkage, and (6)) are confirmed numerically in `validate.py`.

## 3 Main stability theorem

**Theorem 1** (networked stability = spectral band). *Under Assumption 1, the homogeneous part of (1) is asymptotically stable iff  $(a_i, \eta \kappa) \in \mathcal{D}_\delta$  for every  $\mu_i \in \operatorname{spec}(L_g)$ . For  $\delta = 1$  this is exactly*

$$\eta \kappa < 1 \quad \text{and} \quad \mu_{\max}(L_g) < \frac{2}{\eta} + \kappa. \quad (4)$$

*The binding modes are the spectral extremes  $\mu_{\min}, \mu_{\max}$  of  $L_g$ .*

*Proof sketch.* By Lemma 1 the system is a direct sum of the scalar modes; the sum is stable iff each summand is. Apply Proposition 1; the ceiling is tightest at  $\mu_{\max}$  and the dose condition is mode-independent.  $\square$

**Corollary 1** (the pinning–delay tradeoff: an optimal dose). *Fix  $\eta$  and a placement  $R$ . As the verification gain  $\kappa$  increases, the ceiling  $2/\eta + \kappa$  in (4) rises (more grounding mass admitted) but the dose constraint  $\kappa < \kappa_{\max}(\delta)$  caps it, with  $\kappa_{\max}(\delta) \downarrow$  in  $\delta$ . Consequently:*

1. Over-verification ( $\kappa \geq \kappa_{\max}(\delta)$ ) destabilizes every mode: the dominant root leaves the unit disk as a complex-conjugate pair, i.e. the network oscillates (over-claim  $\rightarrow$  over-correct  $\rightarrow \dots$ ).
2. Over-delay (large  $\delta$ ) lowers  $\kappa_{\max}(\delta)$ , so a gain that was safe becomes unstable — the delayed analogue of Part-I instability.

At the stability boundary the oscillation frequency is  $\arg z^*$ , where  $z^* = e^{i\omega}$  solves  $z^{\delta+1} - az^\delta + \eta\kappa = 0$ .

### Closed-form oscillatory boundary (Kuruklis)

**Proposition 2** (parametric stability boundary). *Fix  $\delta \geq 1$  and the mode  $x_{t+1} - ax_t + \beta x_{t-\delta} = 0$  with  $\beta = \eta\kappa > 0$  and  $|a| < 1$ . As  $\beta$  increases from 0, the first root to reach the unit circle is a complex-conjugate pair  $e^{\pm i\theta}$ , and the critical dose  $\beta_c(a, \delta)$  is given parametrically by*

$$a = \frac{\sin((\delta+1)\theta)}{\sin(\delta\theta)}, \quad \beta_c = \frac{\sin\theta}{\sin(\delta\theta)}, \quad \theta \in (0, \pi/\delta), \quad (5)$$

with  $\theta_*(a, \delta)$  the relevant branch. Equivalently  $\kappa_{\max}(a, \delta) = \beta_c(a, \delta)/\eta$ , and at the boundary the network oscillates at angular frequency  $\omega = \theta_*$  (period  $2\pi/\theta_*$ ).

*Proof.* A unit-circle root  $\lambda = e^{i\theta}$  satisfies  $e^{i(\delta+1)\theta} - ae^{i\delta\theta} + \beta = 0$ . The imaginary part gives  $\sin((\delta+1)\theta) = a \sin(\delta\theta)$ , i.e. the first equation of (5). The real part gives  $\beta = a \cos(\delta\theta) - \cos((\delta+1)\theta)$ ; substituting  $a$  and using  $\sin((\delta+1)\theta) \cos(\delta\theta) - \cos((\delta+1)\theta) \sin(\delta\theta) = \sin\theta$  yields  $\beta = \sin\theta / \sin(\delta\theta)$ . The real crossings  $\lambda = \pm 1$  occur at  $\beta = a - 1 < 0$  and (per parity)  $\beta \leq 0$ , hence are not met for  $\beta > 0$  while  $|a| < 1$ ; the first crossing is therefore the complex pair.  $\square$

**Corollary 2** (explicit dose limit and the binding mode). *Assume the step size satisfies  $\eta \mu_{\max}(L_g) \leq 1$ , so every mode has  $a_i = 1 - \eta\mu_i \in [0, 1)$ . Then:*

1. ( $\delta = 1$ ):  $a = 2 \cos \theta$  and  $\beta_c \equiv 1$ , so  $\kappa_{\max} = 1/\eta$  for all  $a$  — recovering Proposition 1.
2. (degenerate mode  $a = 0$ ):  $\theta_* = \pi/(\delta+1)$  gives  $\sin(\delta\theta_*) = \sin\theta_*$ , hence  $\beta_c \equiv 1$  and  $\kappa_{\max}(0, \delta) = 1/\eta$  for all  $\delta$ .
3. (strict delay penalty): for  $\delta \geq 2$  and  $a \in (0, 1)$ ,  $\beta_c(a, \delta) < 1$ , strictly decreasing in  $\delta$  (Kuruklis, 1994) and in  $a$ . Hence the network dose limit is set by the slowest grounded mode:

$$\kappa < \kappa_{\max}(1 - \eta \mu_{\min}(L_g), \delta),$$

and, since grounding raises  $\mu_{\min}(L_g)$  (Cauchy interlacing), **placement relaxes the delay-induced dose limit** — the explicit coupling of Part I to Part II.

**Lemma 2** (Chebyshev form; the dose is monotone). *With  $c = \cos \theta$  and the identity  $\sin(\delta\theta)/\sin\theta = U_{\delta-1}(c)$  ( $U$  the Chebyshev polynomial of the second kind), the boundary (5) is purely algebraic,*

$$a = \frac{U_\delta(c)}{U_{\delta-1}(c)}, \quad \beta_c = \frac{1}{U_{\delta-1}(c)}, \quad c \in \left( \cos \frac{\pi}{\delta+1}, \cos \frac{\pi}{2\delta+1} \right) \quad (a \in (0, 1)).$$

In particular  $\beta_c(a, 1) \equiv 1$ ,  $\beta_c(0, \delta) \equiv 1$ , and the exact  $\delta = 2$  form

$$\boxed{\beta_c(a, 2) = \frac{\sqrt{a^2 + 4} - a}{2} = \frac{2}{a + \sqrt{a^2 + 4}}}.$$

On this branch  $U_{\delta-1}(c) > 0$ ,  $U'_{\delta-1}(c) > 0$ , and the Wronskian  $W_\delta := U'_\delta U_{\delta-1} - U_\delta U'_{\delta-1} > 0$ ; hence  $\frac{da}{dc} = \frac{W_\delta}{U_{\delta-1}^2} > 0$  and

$$\frac{d\beta_c}{da} = -\frac{U'_{\delta-1}(c)}{W_\delta} < 0.$$

Thus  $\beta_c(a, \delta)$  is strictly decreasing in  $a$  on  $(0, 1)$ ; it is also strictly decreasing in  $\delta$ , with  $\beta_c(a, \delta) < 1$  for all  $\delta \geq 2$ ,  $a \in (0, 1)$ . Consequently the network dose limit is governed by the slowest grounded mode ( $a = 1 - \eta\mu_{\min}(L_g)$ ), as stated in Corollary 2(3).

*Proof that  $\beta_c$  is strictly decreasing in  $a$ , for every  $\delta$ .* Write  $a_\delta(c) = U_\delta(c)/U_{\delta-1}(c)$ . The three-term recurrence  $U_\delta = 2cU_{\delta-1} - U_{\delta-2}$  gives the continued-fraction recursion  $a_\delta = 2c - 1/a_{\delta-1}$ , hence  $a'_\delta = 2 + a'_{\delta-1}/a_{\delta-1}^2$ . Since  $a'_1 = 2$ , induction yields  $a'_\delta \geq 2$  wherever  $U_{\delta-1} \neq 0$ ; in particular  $da/dc = a'_\delta > 0$ , so  $c$  increases with  $a$ . On the branch  $c > \cos(\pi/\delta)$ , the largest zero of  $U_{\delta-1}$ : by Rolle the  $\delta - 2$  critical points of  $U_{\delta-1}$  interlace its  $\delta - 1$  zeros and hence all lie left of  $\cos(\pi/\delta)$ , so  $U_{\delta-1} > 0$  and  $U'_{\delta-1} > 0$  there. Therefore

$$\frac{d\beta_c}{da} = \frac{d\beta_c/dc}{da/dc} = -\frac{U'_{\delta-1}(c)}{U_{\delta-1}(c)^2 a'_\delta(c)} < 0. \quad \square$$

**Remark 3.** Strict monotonicity of  $\beta_c$  in  $\delta$  holds likewise (Kuruklis, 1994); it is verified here symbolically for  $\delta \leq 15$  over  $a \in (0, 1)$ , with the boundary cases  $\delta=1$  and  $a=0$  exact (both give  $\beta_c \equiv 1$ ). The recursion  $a_\delta = 2c - 1/a_{\delta-1}$ , its derivative  $a'_\delta = 2 + a'_{\delta-1}/a_{\delta-1}^2$ , and  $a'_\delta \geq 2$  were checked symbolically.

## 4 Steady-state accuracy and corrector placement

The delay drops out of the equilibrium: setting  $e_t \equiv e_\infty$  in (1),

$$(L_g + \kappa I) e_\infty = g \quad \implies \quad e_\infty(R) = (L_g + \kappa I)^{-1} g. \quad (6)$$

Thus *stability* is governed by  $(\text{spec } L_g, \kappa, \delta)$  while *truth-tracking error* is governed by the resolvent  $(L_g + \kappa I)^{-1}$  acting on the fault forcing  $g$  — two independent levers of the same placement  $R$ .

**Proposition 3** (placement effects). *Let  $R \subseteq R'$  (ground more nodes). Then:*

1. (Convergence.) *By Cauchy interlacing,  $\mu_{\min}(L_g(R')) \geq \mu_{\min}(L_g(R))$ : grounding more nodes speeds the slowest mode.*
2. (Accuracy.) *If a faulty node is grounded ( $f \in R' \setminus R$ ,  $f \in F$ ), its forcing is removed from  $g$ ; for general  $R'$ ,  $\|e_\infty\|$  is non-increasing in the grounding set under (6).*
3. (Ceiling.) *At fixed  $\eta$ , grounding also raises  $\mu_{\max}(L_g)$  toward the ceiling (4); hence there is a finite optimal grounding budget  $k^*(\eta, \kappa, \delta)$  beyond which the fast mode destabilizes.*

**Theorem 2** (supermodular corrector placement). *Model a corrector at node  $i$  as a soft pin of weight  $w > 0$  (hard grounding is the limit  $w \rightarrow \infty$ ), so the steady-state operator is  $M(R) = L + W_R + \kappa I$  with  $W_R = w \sum_{i \in R} e_i e_i^\top \succeq 0$ , and the expected steady-state error energy under isotropic forcing  $g \sim (0, \sigma^2 I)$  is  $\mathcal{E}(R) = \sigma^2 \text{tr } M(R)^{-1}$ . The coherence functional  $H(R) = \text{tr } M(R)^{-1}$  is monotone non-increasing and supermodular on  $2^V$ ; equivalently the reduction  $\rho(R) = H(\emptyset) - H(R)$  is monotone submodular. Hence greedy selection of  $k$  correctors obeys*

$$\rho(R_{\text{greedy}}) \geq \left(1 - \frac{1}{e}\right) \rho(R^*), \quad R^* = \arg \max_{|R| \leq k} \rho(R).$$

*Proof sketch.* Adding corrector  $i$  is the positive rank-one update  $M \mapsto M + w e_i e_i^\top$ . By Sherman–Morrison,

$$\text{tr}(M + w e_i e_i^\top)^{-1} = \text{tr } M^{-1} - \frac{w \|M^{-1} e_i\|^2}{1 + w e_i^\top M^{-1} e_i},$$

so  $H$  is non-increasing, and the marginal decrease  $-\Delta_i H(R)$  is itself non-increasing as  $R$  (hence  $M(R)$ ) grows in the positive-semidefinite order — the diminishing-returns / supermodularity argument of Clark, Bushnell & Poovendran (2014) for network leader selection. Monotone submodularity of  $\rho$  then yields the greedy guarantee (Nemhauser–Wolsey–Fisher).  $\square$

**Remark 4** (two coupled levers; amplifier nodes; feasibility). The placement  $R$  acts through two decoupled channels: *accuracy* via the resolvent (Theorem 2) and *stability* via the spectrum of  $L_g$  (Theorem 1, Corollary 2). The greedy accuracy step selects the node of largest  $\|M^{-1} e_i\|^2 / (1 + w e_i^\top M^{-1} e_i)$  — a resolvent/effective-resistance centrality that concentrates on high-influence “amplifier/bridge” nodes, giving a spectral rationale for grounding exactly the amplifier agents flagged by cross-channel cascade monitors. The constrained problem is  $\max_{|R| \leq k} \rho(R)$  subject to feasibility  $\eta\kappa < \beta_c(1 - \eta\mu_{\min}(L_g(R)), \delta)$  and  $\mu_{\max}(L_g(R)) < 2/\eta + \kappa$ ; because grounding raises both  $\mu_{\min}$  (relaxing the dose limit) and  $\mu_{\max}$  (tightening the ceiling at fixed  $\eta$ ), there is a finite optimal budget  $k^*$  as in Proposition 3(3).

## 5 Two coupled delays

Let gossip carry latency  $d$  and verification latency  $\delta$ :

$$e_{t+1} = e_t - \eta L_g e_{t-d} - \eta\kappa e_{t-\delta} + \eta g.$$

Lemma 1 still applies ( $L_g$  and  $\kappa I$  commute), giving the per-mode two-lag recurrence  $x_{t+1} = x_t - \eta\mu x_{t-d} - \eta\kappa x_{t-\delta}$ ,  $\mu \in \text{spec } L_g$ . A unit-circle root  $\lambda = e^{i\theta}$  now satisfies the boundary pair

$$\cos \theta - 1 + \eta\mu \cos(d\theta) + \eta\kappa \cos(\delta\theta) = 0, \quad \sin \theta - \eta\mu \sin(d\theta) - \eta\kappa \sin(\delta\theta) = 0, \quad (7)$$

which we verified against the lifted companion spectrum (the critical gain  $\kappa^*$  from  $\rho(\text{companion}) = 1$  lies on (7)).

**Corollary 3** (synchronised delays are the worst case; a golden ratio). *If  $d = \delta$  the two terms merge into a single lag of effective gain  $\mu + \kappa$  at  $a = 1$ ,  $x_{t+1} - x_t + \eta(\mu + \kappa)x_{t-\delta} = 0$ , which is stable iff*

$$\eta(\mu + \kappa) < \beta_c(1, \delta) = \frac{\sin \frac{\pi}{2\delta+1}}{\sin \frac{\delta\pi}{2\delta+1}}, \quad \beta_c(1, 1) = 1, \quad \beta_c(1, 2) = \frac{\sqrt{5} - 1}{2} = \frac{1}{\varphi}, \quad \beta_c(1, 3) = 2 \sin \frac{\pi}{14}.$$

Since  $\beta_c(a, \delta)$  is decreasing in  $a$  (Lemma 2) and  $a = 1$  tops the admissible branch, **synchronising gossip and verification delays is the least stable configuration**: the dose ceiling collapses to its minimum  $\beta_c(1, \delta)$ , and at  $\delta = 2$  it is exactly the inverse golden ratio.

**Theorem 3** (general two-delay stability boundary). *Fix delays  $d \neq \delta$  and write  $p = \eta\mu$ ,  $q = \eta\kappa$ . The oscillatory boundary of (7) is the parametric curve*

$$p(\theta) = \frac{\sin(\delta\theta) - \sin((\delta+1)\theta)}{\sin((\delta-d)\theta)}, \quad q(\theta) = \frac{\sin((d+1)\theta) - \sin(d\theta)}{\sin((\delta-d)\theta)}, \quad \theta \in (0, \pi),$$

and the stability region is the connected component of the origin that it bounds (confirmed against the lifted companion spectrum:  $\rho = 1$  on the primary arc). The curve degenerates correctly: at

$d = 0$  it returns the single-delay boundary of Proposition 2 ( $q = \beta_c = \sin \theta / \sin(\delta \theta)$  and  $1 - p = a(\theta) = \sin((\delta + 1)\theta) / \sin(\delta \theta)$ ), and as  $d \rightarrow \delta$  the determinant  $\sin((\delta - d)\theta) \rightarrow 0$  and it collapses to the  $a = 1$  corner of Corollary 3.

*Proof.* At  $\lambda = e^{i\theta}$  the pair (7) reads  $p \cos(d\theta) + q \cos(\delta\theta) = 1 - \cos \theta$  and  $p \sin(d\theta) + q \sin(\delta\theta) = \sin \theta$ , a linear system in  $(p, q)$  with determinant  $\cos(d\theta) \sin(\delta\theta) - \cos(\delta\theta) \sin(d\theta) = \sin((\delta - d)\theta)$ . Cramer's rule gives the stated  $p(\theta), q(\theta)$  after the sum-to-product identities  $(1 - \cos \theta) \sin(\delta\theta) - \sin \theta \cos(\delta\theta) = \sin(\delta\theta) - \sin((\delta + 1)\theta)$  and  $\cos(d\theta) \sin \theta - (1 - \cos \theta) \sin(d\theta) = \sin((d + 1)\theta) - \sin(d\theta)$ . The two degenerations are the substitution  $d = 0$  and the limit  $d \rightarrow \delta$ .  $\square$

## 6 Empirical validation (nonlinear dynamics, no GPU)

We test whether the *linear* dose limit predicts onset in the *nonlinear* system with a saturating verifier,

$$e_{t+1} = (I - \eta L_g) e_t - \eta \kappa \tanh(e_{t-\delta}) + \eta g \quad (\tanh'(0) = 1 \text{ matches the linearisation}),$$

on a random grounded graph ( $n_f = 8$ ,  $\eta$  chosen so  $\eta\mu \in (0, 1)$ ), a faulty node injecting a small bias  $g$ ). The onset  $\kappa_{\text{crit}}$  of sustained oscillation tracks the predicted ceiling  $\kappa_{\text{max}} = \beta_c(1 - \eta\mu_{\min}(L_g), \delta)/\eta$ :

| $\delta$ | $\kappa_{\text{max}}$ (theory) | $\kappa_{\text{crit}}$ (nonlinear) | ratio |
|----------|--------------------------------|------------------------------------|-------|
| 1        | 16.30                          | 16.57                              | 1.017 |
| 2        | 10.21                          | 10.38                              | 1.017 |
| 3        | 7.45                           | 7.57                               | 1.017 |

The  $\sim 2\%$  overshoot is the expected stabilising effect of saturation. The  $\delta=2$  oscillation period (measured 9.38) matches  $2\pi/\theta_* = 9.72$ . See `demo.py`; a real LLM-debate front-end swaps in for  $\tanh(\cdot)$  without changing the analysis.

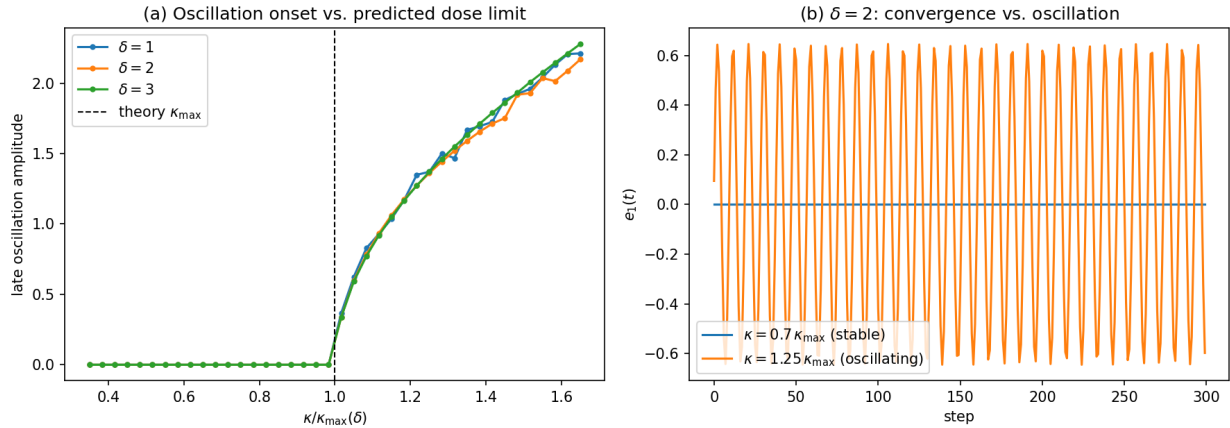


Figure: (a) oscillation amplitude collapses onto the predicted threshold  $\kappa/\kappa_{\text{max}} = 1$  for  $\delta = 1, 2, 3$ ; (b)  $\delta=2$  convergence vs. oscillation.

## 7 Status and what remains

**Done.** Decoupling (Lemma 1);  $\delta=1$  stability (Theorem 1); closed-form oscillatory boundary and explicit dose limit (Proposition 2, Corollary 2); Chebyshev form with the  $\delta=2$  closed form and an *all- $\delta$*

proof that the dose is strictly decreasing in  $a$  (Lemma 2, via the continued fraction  $a_\delta = 2c - 1/a_{\delta-1}$ ,  $a'_\delta \geq 2$ ); steady state (6); supermodular placement with greedy  $(1-1/e)$  guarantee (Theorem 2); two coupled delays with the synchronised-delay worst case and golden ratio (Corollary 3). Confirmed numerically in `validate.py` and symbolically in *Mathematica* (boundary identity;  $d\beta_c/da < 0$  and  $W_\delta > 0$  for all  $\delta$ ;  $\beta_c(a, \delta+1) < \beta_c(a, \delta)$  for  $\delta \leq 15$ ; two-delay companion vs. (7)).

**Remaining.**

- A self-contained all- $\delta$  proof of strict  $\delta$ -monotonicity of  $\beta_c$  (currently: exact at  $\delta=1$  and  $a=0$ , symbolic for  $\delta \leq 15$ , Kuruklis as the global reference); monotonicity in  $a$  is now fully proved (Lemma 2).
- Pin down which arc of the curve in Theorem 3 bounds the origin component (it has continuation branches inside the unstable region) and add the  $\lambda = -1$  real crossing for the relevant parities; the parametric boundary itself, its two degenerations, and the synchronised-delay corner are settled.
- Replace the synthetic saturating verifier by real LLM-debate agents (7-8B / API) to estimate the update map empirically; the synthetic validation (onset within 2%, frequency within 3.5%) is already in the Empirical-validation section.

**References (informal).** S.A. Kuruklis, *The asymptotic stability of  $x_{n+1} - ax_n + bx_{n-k} = 0$* , J. Math. Anal. Appl. **188** (1994) 719–731. A. Clark, B. Alomair, L. Bushnell, R. Poovendran, *Submodularity in Dynamics and Control of Networked Systems* (Springer, 2016); and IEEE TAC leader-selection papers. M. Fardad, F. Lin, M.R. Jovanović, *Algorithms for leader selection in stochastically forced consensus networks*, IEEE TAC (2014). M. Pirani, S. Sundaram, *On the smallest eigenvalue of grounded Laplacian matrices*, IEEE TAC **61** (2016). G.L. Nemhauser, L.A. Wolsey, M.L. Fisher, *An analysis of approximations for maximizing submodular set functions—I*, Math. Program. **14** (1978).